

FLOODING *IN VIVO* FOR A CASE OF VERMIPHOBIA*

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Summary—A young woman with an incapacitating fear of worms and snakes, who was refractory to psychoanalysis and to desensitization, was treated by *in vivo* flooding. Live worms and a European grass snake were used as the phobic stimuli in six therapy sessions ranging from 3 hr to $\frac{1}{2}$ hr in duration. After the 10 hr of therapy the patient showed a marked reduction in phobic behavior.

THERE have been several reports of *in vivo* desensitization of animal phobias (e.g. Freeman and Kendrick, 1960; Murphy, 1964). The treatment of the case to be described here was considerably different in detail.

CASE HISTORY

I.L. was a 29-yr-old English woman who had been both an inpatient and an outpatient at the Maudsley Hospital, London for several years.

She claimed to have suffered since 4 yr of age, from a "fear of worms, snakes and anything that wriggles". She was unable to walk out in the rain without great difficulty for fear of seeing a worm on the ground. Even in dry weather, she constantly glanced at the ground searching for worms. She reported a recurrent nightmare where "worms were crawling all over my body". She did not eat fish, celery, watercress, or lettuce for fear that these foods were contaminated with worms. If she found herself standing near a worm or was touched by someone who handled a worm, she felt a need to wash until she no longer felt contaminated. Small, pink worms wriggling in moist mud were most fearful to I.L., but dry worms on a table top also caused some anxiety. She also expressed a fear of

snakes which was not as severe as that of worms.

In none of the interviews could the patient offer any rational explanation for her fears. She described herself as being gripped by an overpowering anxiety when confronted with the phobic object. Her symptoms included perspiration, palpitation, tachypnea, nausea, and a sense of tightness in her chest. She adequately fitted Frazier and Carr's (1970) description of a phobic personality. Her case was strikingly similar to that described by Murphy (1964).

TREATMENT

The patient agreed to remain in the therapeutic environment (e.g. room, garden) throughout the entire session and was assured that she would not be forced to do anything she felt absolutely incapable of doing. The therapy was divided into three phases:

- (1) *In vivo* flooding with six worms in a small box filled with moist mud.
- (2) Digging in the garden for worms with a shovel.
- (3) *In vivo* flooding with a short European grass snake (*Natrix natrix*).

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After modelling by the therapist, the patient was asked to stand in front of the box of worms at a distance of 3 m and continue to look at them. Jumping back was discouraged but the venting of anxiety by other methods such as screaming was permitted. Thus, the patient was kept in the phobic situation and could not receive the reinforcement of anxiety reduction that would ensue on running away from the worms. Actually, she gradually moved closer. After 1½ hr of continuous exposure she stood within 1 m of the box and after an additional 1½ hr was able to touch the edge of the box, but neither the mud nor the worms. The second 3-hr session of the first phase began with the patient standing one meter from the box. Within ½ hr she returned to the point where she had left off at the previous session. After two more hours she touched the wriggling worms in the box, and in the final ½ hr rapidly progressed through holding the wriggling worms in her hand, letting them wriggle on her thigh and running them through her hair. Since the patient reported nearly negligible anxiety to live worms at this stage, the second phase of the therapy was commenced.

I.L. was instructed to search for worms in the ground with her shovel and remove them from the wet earth into a cardboard box. All aspects were modelled for her initially by the therapist. After digging in the hospital garden for ½ hr the patient encountered a worm at a depth of 18 in. For 1 hr the patient transferred the worm from the ground to the box and back again. This procedure was repeated in a second session lasting ½ hr, and the second phase of therapy therefore lasted for a total of 2 hr. It was considered to be concluded when she could remove the worms from the ground without fear.

The third phase of therapy was the same as the first, but with the box of worms replaced by a grass snake and cloth sack, initially at a distance of 3 m. After 20 min I.L. sat with the snake in the sack on her lap, and by 1½ hr she calmly held the snake in her hand. In the second session of the third phase, the sixth and final session of her therapy, she spent ½ hr handling

the snake and concluded by placing it in the sack and fastening the string. Since the patient could handle both worms and snakes, the therapy was terminated.

RESULTS

Griffiths and Joy (1971) having shown Walk's 10 point rating scale (1956) to be a reliable method of monitoring phobic anxiety, this method was used with I.L. Changes both within and between sessions were noted. One week after the therapy was concluded the patient was markedly improved as shown in Table 1. She claimed that she was able to sleep with an open box of worms in her bedroom and had experienced no nightmares that week. I saw her walk in the hospital garden with her head erect. Six months later, her gains were found, by and large, to have been maintained (Table 1).

TABLE 1. PATIENT SUBJECTIVE RATINGS OF FEAR ON 10 POINT RATING SCALE*

Event	Pre-treatment	1-week Post-treatment	6-month Follow-up
1. walking in rain	9	1	1
2. glancing at ground	9	0	0
3. staring at picture of worm	7	0	0
4. touching dead worm	10	1	2
5. touching live wriggling worm	10	2	4
6. touching live snake	9	1	0
7. fear of contamination by worms	8	0	1

*0 = calm, 10 = panic.

It is worth noting that the patient was taking tranquilizing medication between therapy sessions; but the same medication administered for several months previously had had no effect on her vermiphobia.

Flooding is not without hazard. It is an exhausting and tension-ridden process, which must be carefully explained to the patient in advance.

While *in vivo* flooding generally involves the actual phobic object in the therapy session, methods have been devised to simulate certain life situations (e.g. tape recordings of thunderstorms for treatment of thunder phobics).

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